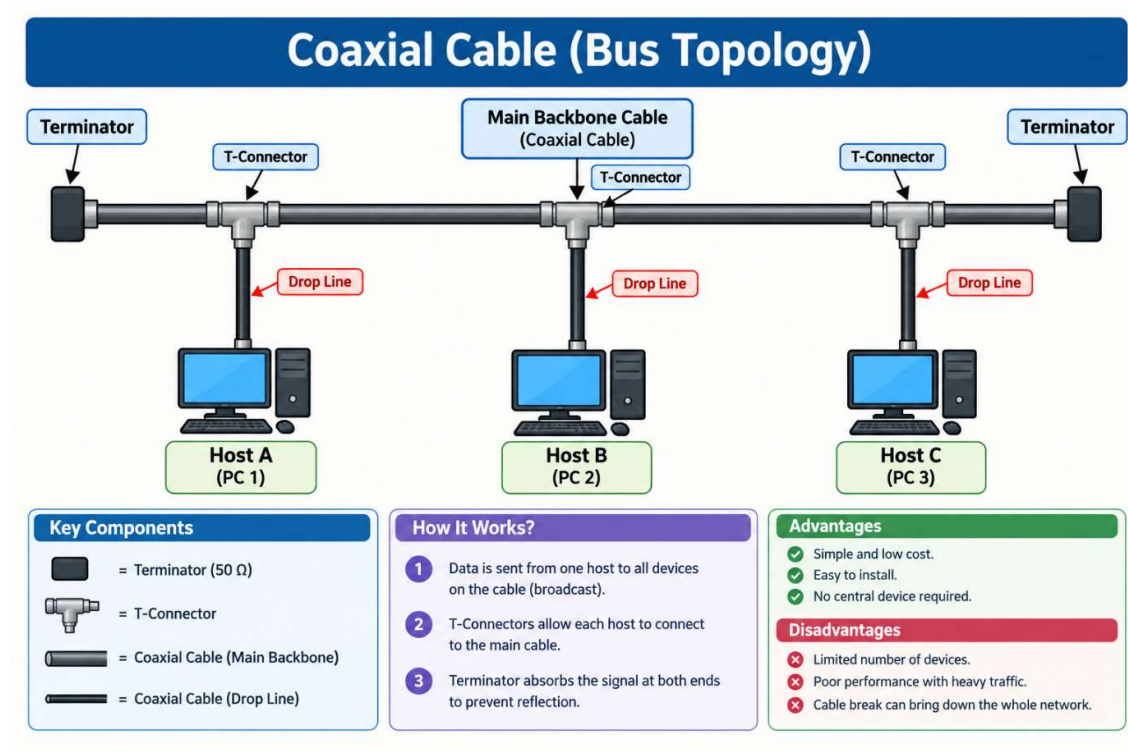


Task 1:

Bus Topology Architecture & Real-World Application

Objective: Design a structural architectural diagram of a Bus Topology, label all core hardware components, and identify a practical real-world implementation.

1. Bus Topology Architectural Diagram:



2. Labeled Component Description:

- **Backbone Cable:** The central shared communication line (typically coaxial cable) that carries data signals across all connected devices.
 - **Terminators:** Absorbing devices placed at both physical ends of the backbone cable to prevent signal bouncing and packet reflections.
 - **Drop Lines:** Small extension cables connecting individual node network interface cards (NICs) to the main backbone cable.
 - **T-Connectors (BNC Connectors):** Physical couplers used to splice drop lines into the main backbone transmission line.
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3. Real-World Application Example:

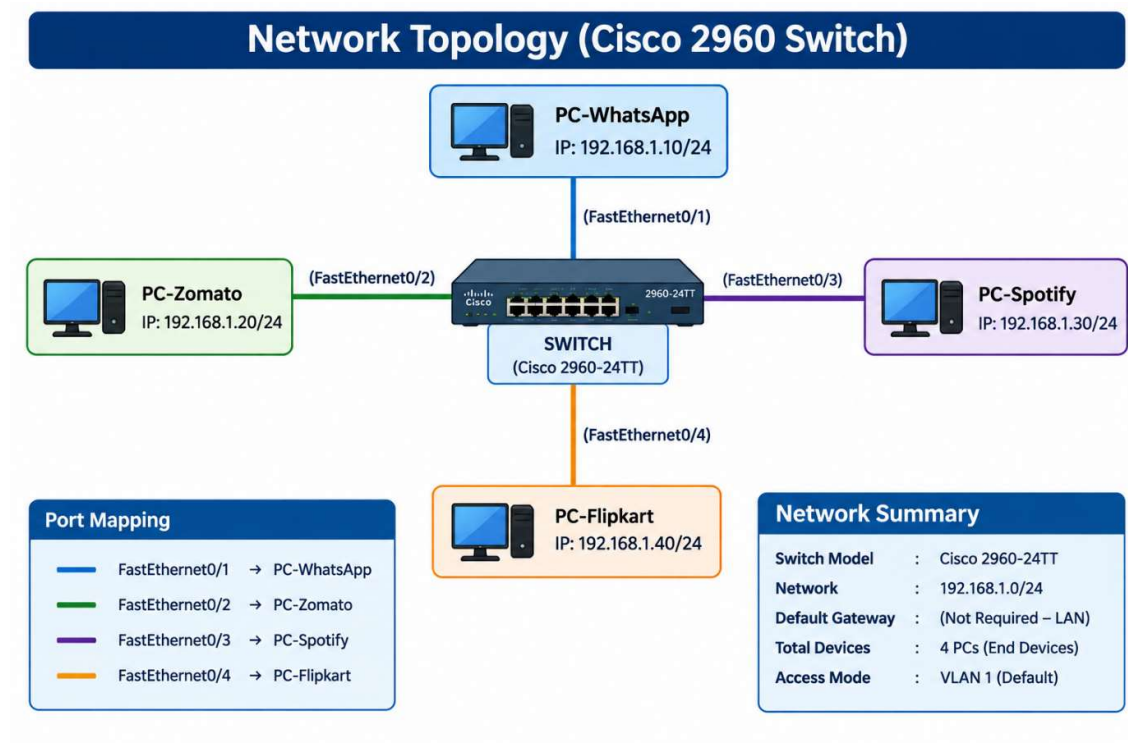
- **Legacy Ethernet Networks (10Base2 / 10Base5) & Small Office LANs:** Bus topology was historically used in early local area networks (LANs) connecting a small group of workstation computers in a single room via thick coaxial cabling without requiring an active central hub or switch.
- **Modern Embedded Systems (CAN Bus):** Used in automotive **Controller Area Network (CAN Bus)** systems where multiple sensors, microcontrollers, and electronic control units (ECUs) inside a vehicle communicate over a single shared bus line.

Task 2:

Cisco Packet Tracer Star Topology Simulation

Objective: Design and simulate a centralized Star Topology in Cisco Packet Tracer consisting of one central network switch connected to four endpoints named after popular applications (WhatsApp, Zomato, Spotify, and Flipkart).

1. Visual Network Topology Diagram:



2. Device Configuration & Port Assignment Table:

Conn ected Devic e Name	Dev ice Typ e	Switch Interfa ce Port	Assign ed IP Addres s	Subnet Mask	Conne ction Cable Type
Centr al Switc h	Cis co 296 0 Swi tch	N/A (Centr al Node)	<i>N/A</i>	<i>N/A</i>	N/A
PC- What sApp	End Dev ice (PC)	FastEt hernet 0/1	192.16 8.1.10	255.255 .255.0	Coppe r Straigh t- Throug h
PC- Zomat o	End Dev ice	FastEt hernet 0/2	192.16 8.1.20	255.255 .255.0	Coppe r Straigh t-

Conn ected Devic e Name	Dev ice Typ e	Switch Interfa ce Port	Assign ed IP Addres s	Subnet Mask	Conne ction Cable Type
	(PC)				Throug h
PC- Spotif y	End Dev ice (PC)	FastEt hernet 0/3	192.16 8.1.30	255.255 .255.0	Coppe r Straigh t- Throug h
PC- Flipka rt	End Dev ice (PC)	FastEt hernet 0/4	192.16 8.1.40	255.255 .255.0	Coppe r Straigh t- Throug h

3. Cisco Packet Tracer Steps & Screenshot Placeholder:

1. **Place Central Node:** Selected Network Devices > Switches > 2960 and dragged it to the center workspace.
2. **Place End Devices:** Dragged 4 PC endpoints and renamed their display labels to **PC-WhatsApp**, **PC-Zomato**, **PC-Spotify**, and **PC-Flipkart**.
3. **Cabling:** Connected each PC's FastEthernet0 port to the Switch's FastEthernet0/1 through FastEthernet0/4 ports using **Copper Straight-Through Cables**.
4. **IP Assignment:** Configured static IP addresses (192.168.1.10 to 192.168.1.40 with /24 subnet masks) on each PC's Desktop interface.
5. **Verification:** Tested connectivity by pinging 192.168.1.20 from **PC-WhatsApp**, confirming successful Layer 2 switching.

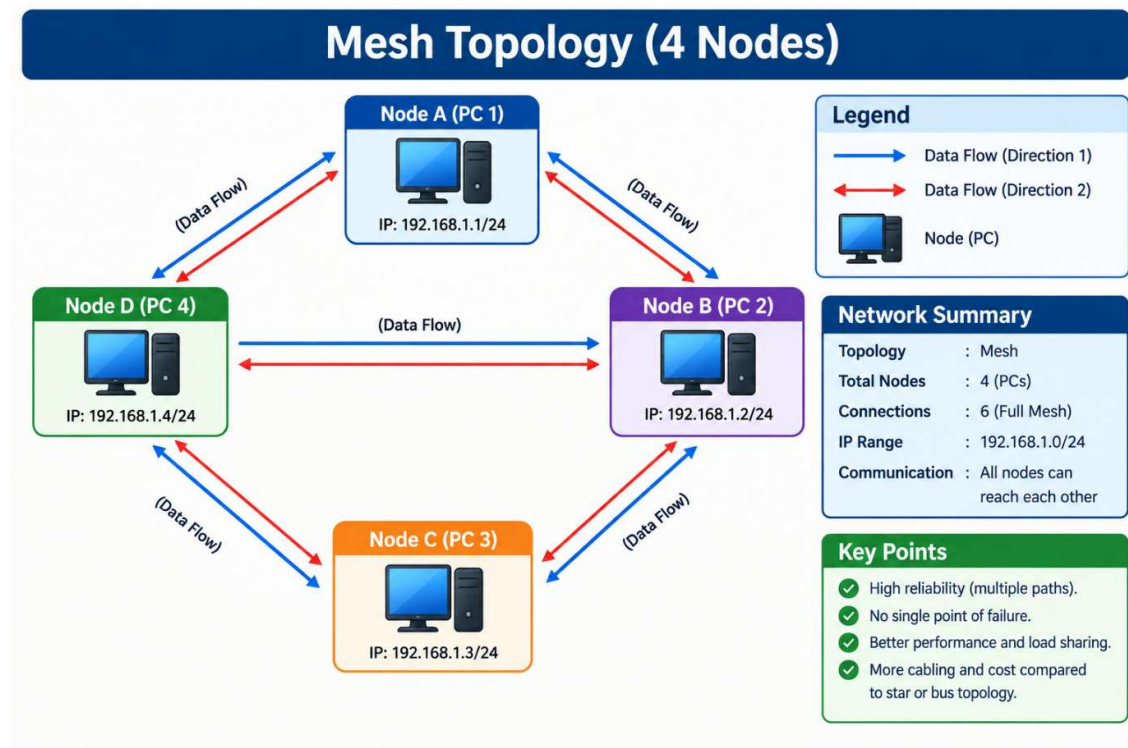
[Insert Cisco Packet Tracer Star Topology Screenshot Here
]

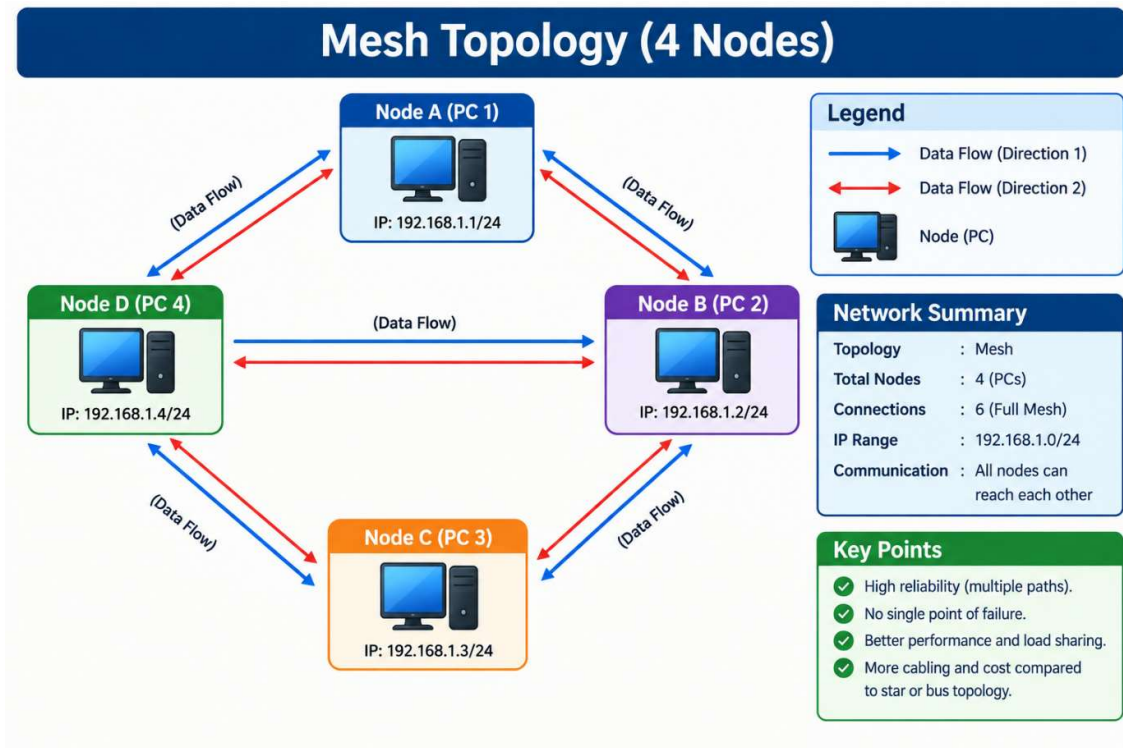
Task 3:

Ring Topology Architecture & Data Flow Analysis

Objective: Create an architectural diagram of a Ring Topology showing directional data flow arrows and explain the token-passing data transmission mechanism in 2–3 concise sentences.

1. Ring Topology Architectural Diagram:





2. How Data Travels in Ring Topology (2-3 Sentences Explanation):

In a ring topology setup, data packets travel sequentially in one continuous direction (unidirectional ring) from one node to the next using a **token-passing protocol**. Every intermediate device acts as a repeater that receives the packet, inspects the destination MAC/IP address, and regenerates the signal to pass it forward along the loop until it reaches the intended recipient node. Once the recipient receives the data, an acknowledgment frame is sent back along the ring to release the token for other devices to transmit.

Task 4:

Comparative Analysis: Mesh Topology vs. Star Topology

Objective: Evaluate and compare Mesh Topology and Star Topology by detailing two key advantages and two disadvantages for each architecture in a structured table format.

Mesh vs. Star Topology Comparison Table:

Network Topology	Advantages (Pros)	Disadvantages (Cons)
Mesh Topology	1. Exceptional Fault Tolerance & Redundancy: Dedicated point-to-point links ensure that if one cable or device fails, data automatically reroutes through alternate links	1. Extremely High Installation & Hardware Cost: Requires an excessive amount of cabling and multiple Network Interface Cards (NICs) per device.

Network Topology	Advantages (Pros)	Disadvantages (Cons)
	without disrupting the network. 2. High Security & Privacy: Direct private links between nodes prevent unauthorized eavesdropping, data interception, or traffic congestion from other devices.	2. Complex Configuration & Maintenance: As the number of nodes grows, the number of required physical links expands exponentially ($N(N - 1)/2$), making setup and troubleshooting highly complex.
Star Topology	1. Easy Setup & Scalability: Simple to install and expand; adding or removing new endpoint nodes	1. Single Point of Failure: The entire network depends entirely on the central device (Switch/Hub); if the

Network Topology	Advantages (Pros)	Disadvantages (Cons)
	requires plugging a single cable into the central switch without taking down the network. 2. Centralized Troubleshooting & Isolation: Faults are easy to locate and isolate; if a single link or host fails, only that specific host loses connection while the rest of the network operates normally.	central switch fails, the whole network collapses. 2. Cable Length & Switch Dependence: Performance is constrained by the processing capacity and port density of the central switch, requiring more total cabling compared to bus or ring topologies.